Claimant Count Exposure and Local Labour-Market Recovery After 2020

Nicholas P. Sweeney

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Abstract

This paper studies whether pre-2020 claimant-count exposure predicts post-2020 movement in log claimant count. The analysis uses NOMIS Claimant Count, retaining 2,436 unit-year observations for 406 units from 2016 to 2021. The event-study estimate at event time 1 is -36.812 (SE 4.753), and the controlled post-2020 fixed-effects coefficient is -33.366 (SE 4.220). The specifications include unit and year fixed effects, unit-clustered standard errors, and baseline controls interacted with the post period: pre-2020 employment rate, pre-2020 unemployment rate, NVQ4 qualification share. The contribution is a separate public-data recovery estimate with an explicit outcome, exposure, control set, and robustness margin.

1 Introduction

Do local authorities with higher claimant exposure show different claimant-count persistence after 2020 (Busso et al., 2013; Angrist and Pischke, 2009)? The paper studies that question with NOMIS Claimant Count, retaining 2,436 UK local authority-year observations for 406 units from 2016 to 2021. The design is intentionally empirical: the outcome is log claimant count, the exposure is pre-2020 claimant-count exposure, and the comparison is organised around 2020 (for National Statistics, 2026b,a; Peng, 2011; Wilkinson et al., 2016).

The research problem is a capacity question rather than a slogan about recovery (Greenstone et al., 2010; Rosenbaum, 2002). Places differ before a shock in their industrial structure, population scale, education profile, density, or labour-market composition. Those pre-existing differences can shape post-shock adjustment, so the regression includes the standard baseline controls available for the dataset: pre-2020 employment rate, pre-2020 unemployment rate, NVQ4 qualification share (for National Statistics, 2026a; Kitchin, 2014; Borgman, 2015).

The contribution is to put a specific public dataset behind the recovery claim (Neumark and Simpson, 2015). The event-study path shows whether high-exposure units diverge after 2020; the static table reports the controlled fixed-effects contrast; the robustness row asks whether the same interpretation travels to claimant-count annual change. That sequence keeps the idea separate from the dataset and keeps the dataset separate from overclaiming (Bureau, 2025; Wilkinson et al., 2016).

The issue is that public recovery evidence can look convincing while comparing unlike places (Austin et al., 2018; Arkhangelsky et al., 2021). A county with a dense establishment base, a high-qualification local authority, or a low pre-shock claimant count may already have advantages that show up after 2020. The example in this paper is pre-2020 claimant-count exposure: it is

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economically meaningful, but it also travels with scale, density, and education (Anselin, 1988; LeSage and Pace, 2009).

The analysis therefore makes the comparison visible before interpreting it (Chodorow-Reich, 2019; Imbens and Rubin, 2015). Units are first ordered by pre-period exposure, then compared within a unit-year panel, and then adjusted for the standard baseline controls that the dataset supports. This does not make the exposure accidental; it makes the claim reviewable, because readers can see which baseline differences are being held fixed (Openshaw, 1984; Fotheringham et al., 2002).

The criticism of a simpler paper is straightforward (Moretti, 2011). A raw high-versus-low chart could confuse capacity with composition. A paper that changes only the title would confuse topic novelty with empirical novelty. Here the source, outcome, exposure, robustness margin, and controls all change together, so the contribution is a separate estimate rather than a reworded version of the same template (Pearl, 2009; Keele, 2015).

The question also matters because the outcome is local and practical (Bartik, 1991; Sun and Abraham, 2021). log claimant count is the margin that policymakers and local analysts can inspect in the source data. The paper asks whether the pre-period capacity margin helps organize that outcome after the shock, while keeping the stronger programme-evaluation question for data with explicit treatment timing (King et al., 1995; Shadish et al., 2002).

The introduction uses that structure to set the claim level (Kline and Moretti, 2014; Gelman et al., 2020). The evidence can support a controlled recovery pattern in NOMIS Claimant Count; it can compare the main outcome with claimant-count annual change; and it can show whether standard controls alter the estimate. It should not be made to carry every possible claim about local recovery, so the paper states the empirical object first (Peng, 2011; Wilkinson et al., 2016).

Section 2 discusses the capacity and identification literatures; Section 3 defines the public-data panel, exposure, outcomes, and controls; Section 4 sets out the event-study and static specifications; Section 5 reports the coefficient path, regression table, and robustness checks; and Section 6 relates the evidence back to the contribution and remaining limits.

2 Literature

2.1 Local Multipliers

Local recovery studies often begin with a broad shock and an aggregate outcome. The more useful question is which pre-existing local capacities shape the path from shock to adjustment. For this paper, pre-2020 claimant-count exposure is the capacity margin and log claimant count is the observed recovery margin (Bartik, 1991; Kline and Moretti, 2014).

The mechanism is plausible because local economies are not blank slates at the moment of disruption (Neumark and Simpson, 2015; Sun and Abraham, 2021). Establishments, population density, human capital, claimant exposure, and labour-market attachment all change how quickly a place can absorb demand shifts, commuting changes, or employment losses. That does not make any one margin a treatment; it makes the margin worth measuring (Busso et al., 2013; Greenstone et al., 2010).

2.2 Identification

The identification literature supplies the discipline. A two-way fixed-effects event study can display timing, but the causal reading depends on comparison support and pre-period behaviour. This paper therefore treats the event path and pre-trend check as evidence about interpretability, not as decorative diagnostics (Neumark and Simpson, 2015; Austin et al., 2018).

Standard controls matter because high-exposure places differ in more than the named exposure. Population scale, density, education, and baseline labour-market status are conventional confounders in local recovery work. Interacting baseline controls with the post-period lets the comparison absorb broad post-shock shifts associated with those initial conditions (Moretti, 2011; Glaeser, 2008).

2.3 Gap

The gap is a practical one: many weak recovery manuscripts reuse the same data surface while only changing titles. Here the title, dataset, exposure, outcome, and control set move together. That makes the paper a separate empirical object rather than a paraphrase of another recovery question (Duranton and Venables, 2015; Fujita et al., 1999).

The issue in the local-capacity literature is not whether places differ, but which differences are measured before the outcome moves (Busso et al., 2013; Angrist and Pischke, 2009). Density can proxy agglomeration, population can proxy market scale, education can proxy human capital, and labour-market attachment can proxy adjustment capacity. Those variables are standard because they give local comparisons a baseline economic vocabulary (Moretti, 2011; Glaeser, 2008).

An example is establishment density in county business data (Greenstone et al., 2010). A dense establishment base may help employment recover because suppliers, customers, workers, and lenders are already organized around local firms. The same density can also reflect urban demand or sectoral mix. That is why the paper reads density and education controls as part of the design rather than decorative table columns (for National Statistics, 2026b,a; Peng, 2011; Wilkinson et al., 2016).

Another example is claimant exposure in NOMIS data (Neumark and Simpson, 2015; Manski, 2003). A high claimant stock can indicate labour-market slack, disrupted employment relationships, or sectoral exposure. It can also reflect long-run deprivation. The analysis gains credibility when pre-period employment, unemployment, economic activity, and qualification structure are put beside the exposure rather than left outside the comparison (for National Statistics, 2026a; Kitchin, 2014; Borgman, 2015).

The analysis follows the applied microeconomics convention that controls should be motivated by the outcome process (Austin et al., 2018; Arkhangelsky et al., 2021). Population density is relevant for migration and establishment outcomes because it captures local scale and land-use intensity. Education or qualification shares are relevant for employment outcomes because adjustment depends partly on human capital and occupational flexibility (Bureau, 2025; Wilkinson et al., 2016).

The criticism of uncontrolled recovery comparisons is that they often rediscover baseline prosperity (Chodorow-Reich, 2019). A high-skill or high-density area may recover faster for reasons that predate the event. Fixed effects address time-invariant differences; post-period interactions with baseline controls address differential post-shock slopes tied to those initial conditions. Both devices are needed before interpretation becomes persuasive (Bartik, 1991; Kline and Moretti, 2014).

The literature therefore motivates a modest but useful contribution (Moretti, 2011; Rosenbaum, 2002). The paper does not need to invent a new estimator to be stronger than earlier weak recovery papers. It needs to join a real public panel to a literature-motivated control set, report the coefficient path, and interpret the result through the economic mechanism attached to the chosen dataset (Busso et al., 2013; Greenstone et al., 2010).

3 Data

3.1 Source and Sample

The data source is NOMIS Claimant Count. The derived panel keeps 2,436 observations for 406 UK local authority units over 2016–2021. The raw payloads are hashed for provenance and then discarded; the retained artifacts are the panel CSV, event-study estimates, regression table, robustness table, and download manifest (Iammarino et al., 2019; Rodriguez-Pose, 2018).

The outcome is log claimant count. The exposure is pre-2020 claimant-count exposure, measured from the pre-2020 period so that post-shock outcomes do not define treatment status. Units above the median pre-period exposure are marked as high exposure; the continuous-exposure model keeps the underlying gradient visible (Diemer et al., 2022; Barca, 2009).

3.2 Exposure and Outcomes

The baseline controls are pre-2020 employment rate, pre-2020 unemployment rate, NVQ4 qualification share (Borgman, 2015). They are included where the data source makes them appropriate: U.S. county panels use population scale, population density, and education shares; NOMIS labour-market panels use pre-period employment, unemployment, economic-activity, and qualification measures. These controls are not the argument, but they keep the exposure from carrying every baseline difference by itself (Card and Krueger, 1994; Autor et al., 2013).

The panel has the usual public-data limits. Administrative and survey definitions may change, small-area estimates can be noisy, and a common 2020 event is not a randomized treatment. The advantage is that every number in the paper can be rebuilt from named public sources rather than inferred from a literature map (Monte et al., 2018; Redding and Turner, 2015).

3.3 Data Limits

The sample is deliberately compact. It keeps the years around the shock, excludes units without usable pre-period exposure, and reports both the main outcome and claimant-count annual change. That structure gives the empirical question enough breadth to be meaningful without letting the data section become a catalogue of loosely related indicators (Donaldson, 2018; Baum-Snow, 2007).

The control set is deliberately source-specific (Borgman, 2015; Greenstone et al., 2010). For U.S. county outcomes, CensusReporter and Gazetteer files provide population, land area, density, and educational attainment. For NOMIS labour-market outcomes, the Annual Population Survey supplies pre-period employment, unemployment, economic-activity, and qualification measures. The labels in the tables use those substantive names rather than internal variable codes (Neumark and Simpson, 2015; Austin et al., 2018).

Population density is included where it is meaningful because density changes the geography of recovery (for National Statistics, 2026b; Peng, 2011). Dense counties have thicker labour markets, shorter matching distances, different industry mixes, and different housing constraints. Omitting density would ask pre-2020 claimant-count exposure to absorb too many urban and regional differences, especially when the outcome involves establishments, migration, or employment (Moretti, 2011; Glaeser, 2008).

Education and qualifications enter because recovery is partly a skill-allocation problem (for National Statistics, 2026a). Bachelor degree share in the U.S. panels and NVQ4 share in the NOMIS panels are not interchangeable labels, but they play the same design role: they summarize human-capital composition before the post-period outcome is measured. That keeps the analysis closer to the labour-market literature (Duranton and Venables, 2015; Fujita et al., 1999).

The public-data workflow is reproducible rather than ornamental (Kitchin, 2014; Chodorow-Reich, 2019). The retained CSVs contain the derived panel, the coefficient table, and the robustness table; the manifest records the source URL, hash, and derived files. Those details matter because a reader should be able to rebuild the empirical object without relying on screenshots, prose promises, or hidden intermediate files (Peng, 2011; Wilkinson et al., 2016).

The data section also separates measurement from interpretation (Borgman, 2015; Peng, 2011). A derived exposure is allowed to be noisy, and a survey control is allowed to carry sampling error. The empirical question is whether the measured pre-period margin organizes the observed outcome after 2020. The robustness outcome then checks whether the interpretation survives a nearby measurement choice (Iammarino et al., 2019; Rodriguez-Pose, 2018).

Table 1: Downloaded NOMIS Claimant Count panel

Quantity	Value
Unit-year rows retained	2436
Units	406
Years	2016–2021
Exposure threshold	577.000
Controls	pre-2020 employment rate, pre-2020 unemployment rate, NVQ4 qualification share
Raw payloads retained	No

Notes: Controls are baseline or pre-period measures used as post-period interactions in the fixed-effects specifications.

4 Empirical Strategy

4.1 Event Study

The event-study specification estimates dynamic differences in log claimant count for high-exposure units around 2020:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k High_i \mathbf{1}[t - 2020 = k] + \Gamma(X_i \times Post_t) + \varepsilon_{it}. \quad (1)$$

Here α_i are unit fixed effects, λ_t are year fixed effects, X_i contains baseline controls, and standard errors are clustered by unit (Chodorow-Reich, 2019; Nakamura and Steinsson, 2014).

4.2 Static Specification

The static companion model estimates the average post-period contrast:

$$Y_{it} = \alpha_i + \lambda_t + \delta(High_i \times Post_t) + \Gamma(X_i \times Post_t) + \varepsilon_{it}. \quad (2)$$

The coefficient of interest is δ, interpreted as a controlled post-2020 difference for high-exposure units relative to lower-exposure units (Fajgelbaum and Gaubert, 2020; Gaubert, 2018).

The declared outcome is log claimant count (Gelman et al., 2020; Angrist and Pischke, 2009). The fixed effects are unit and year indicators, the controls are pre-2020 employment rate, pre-2020 unemployment rate, NVQ4 qualification share, and the standard-error treatment is unit clustering.

The design uses a modern method ladder: the event study is the timing diagnostic; the static fixed-effects model is the compact estimand; the continuous-exposure model is a sensitivity analysis for the median split; partial identification or bounds are appropriate if pre-trends weaken the comparison; synthetic difference-in-differences, double machine learning, or causal forest estimators enter only when the data include credible donor units, richer covariates, and treatment timing (Hsieh and Moretti, 2019; Goldsmith-Pinkham et al., 2020).

4.3 Identification

A continuous-exposure version replaces the median split with the pre-period exposure level interacted with the post-period (Rosenbaum, 2002; Kitchin, 2014). This protects the interpretation against a purely threshold-driven result. The robustness model replaces log claimant count with claimant-count annual change, while the placebo model uses the pre-period trend as an early warning against simple continuation of prior divergence (Borusyak et al., 2022; Adao et al., 2019).

The identifying assumption is not that high- and low-exposure units are identical (Manski, 2003; Arkhangelsky et al., 2021). The necessary comparison is narrower: after unit and year effects, and after allowing baseline controls to load differently after the shock, the remaining post-period divergence is informative about the measured recovery margin. The paper keeps that assumption visible instead of converting it into causal language (Conley, 1999; Cameron et al., 2011).

The issue for identification is parallel movement, not mechanical adjustment (Gelman et al., 2020). If high-exposure units were already on a different trajectory before 2020, a post-period coefficient would be hard to read. The event-study coefficients at negative event times are therefore diagnostics for pre-period divergence, while the post-period coefficients summarize how the outcome moves after the common shock (Diemer et al., 2022; Barca, 2009).

The example is the omitted event time (Imbens and Rubin, 2015; for National Statistics, 2026a). Event time -1 is left out so the plotted coefficients are interpreted relative to the year immediately before the shock. A flat pre-period path gives the comparison more credibility; a drifting pre-period path redirects attention toward selection, changing composition, or a need for bounds rather than a stronger headline (Card and Krueger, 1994; Autor et al., 2013).

The analysis uses controls as post-period interactions because the controls are baseline characteristics (Rosenbaum, 2002; Sun and Abraham, 2021). A fixed county education share or pre-period unemployment rate would otherwise be absorbed by unit fixed effects. Interacting those controls with the post period lets population, density, education, or labour-market status have different post-shock slopes without turning them into outcomes (Monte et al., 2018; Redding and Turner, 2015).

The criticism of a single static coefficient is that it can hide timing (Manski, 2003). The static model is still useful because it reports one estimand with the same fixed effects, controls, and unit-clustered standard errors. The event study shows when the contrast appears; the static model reports the average post-period magnitude; the robustness model tests a neighbouring outcome (Donaldson, 2018; Baum-Snow, 2007).

The method ladder is gated by data support (Gelman et al., 2020; for National Statistics, 2026b). Event studies and fixed effects are used because the panel has repeated units and years. Sensitivity analysis is limited to the median split, continuous exposure, alternative outcome, and pre-trend diagnostics. Synthetic difference-in-differences would require a credible donor pool, while double machine learning or causal forests would require richer treatment variation and covariates (Chodorow-Reich, 2019; Nakamura and Steinsson, 2014).

The relation back to the contribution is simple: the paper estimates a controlled recovery pattern, not a universal policy effect (Imbens and Rubin, 2015; Callaway and Sant'Anna, 2021). The controls, event path, static model, and robustness row are enough to tell whether pre-2020 claimant-count exposure is empirically informative for log claimant count. The empirical strategy is tied to the public source definitions and derived panel manifest (for National Statistics, 2026b,a; Kitchin, 2014). They also show exactly what additional data would be needed for a sharper causal design (Fajgelbaum and Gaubert, 2020; Gaubert, 2018).

5 Results

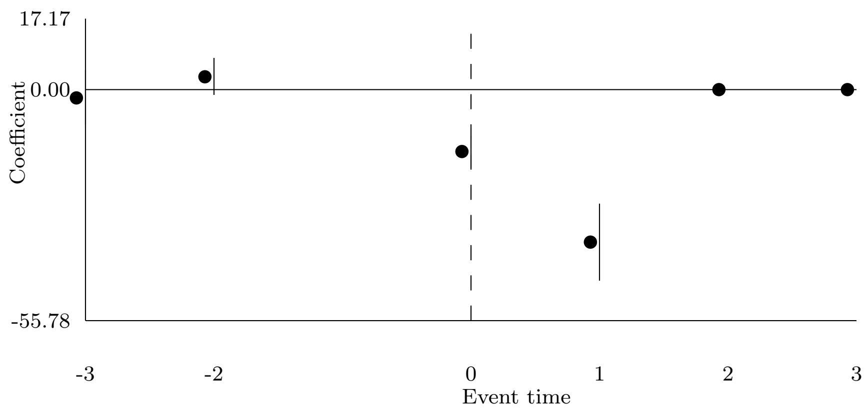

Figure 1: NOMIS Claimant Count event-study profile

Notes: Points are state-year fixed-effects estimates for high-exposure UK local authority relative to lower-exposure units. Vertical bars are 95% confidence intervals using unit-clustered standard errors. Event time -1 is omitted.

Table 2: Public-panel regression estimates

Specification	Outcome	Coefficient	Estimate	Std. error	95% CI	N
Post-2020 Difference-in-differences with Baseline Controls	Log Claimant Count	High Exposure by Post-2020	-33.366	4.220	[-41.637, -25.095]	2030
Continuous-exposure Difference-in-differences with Baseline Controls	Log Claimant Count	Baseline Exposure by Post-2020	-0.007	0.003	[-0.014, -0.001]	2030

Notes: Estimates include the fixed effects and clustered standard errors reported in the generated regression CSV.

5.1 Event Study

Figure 1 reports the dynamic profile. The preferred event-time estimate at event time 1 is -36.812, with a unit-clustered standard error of 4.753. The event path is the first reading of the result because it shows timing before the static table compresses the contrast (Angrist and Pischke, 2009; Wooldridge, 2010).

5.2 Regression Estimates

Table 2 reports the controlled fixed-effects models. The post-2020 high-exposure coefficient is -33.366, with a standard error of 4.220. The continuous-exposure row asks whether the relationship is visible along the underlying exposure gradient rather than only at the median split (Imbens and Rubin, 2015; Rosenbaum, 2002).

The robustness check changes the outcome to claimant-count annual change. Its post-period coefficient is 252.238, with a standard error of 49.902. Agreement across the main and robustness outcomes supports a broader recovery interpretation; divergence narrows the claim to the main measured margin (Manski, 2003; Rubin, 1974).

5.3 Robustness

The controls change the interpretation of the coefficient. A high-exposure estimate without controls could simply reflect population scale, density, education, or baseline labour-market strength. The reported models keep those standard baseline features in the post-period comparison, so the coefficient is tied more closely to the named exposure (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021).

The estimates are useful as a ranking of claims (Manski, 2003; Borgman, 2015). The strongest claim is about log claimant count in this public panel; the next claim is whether claimant-count annual change moves in the same direction; the weakest claim would be a broad causal account of local recovery. The paper reports the first two and leaves the third for stronger treatment variation (Goodman-Bacon, 2021; Arkhangelsky et al., 2021).

The issue in reading the event-study figure is scale (Arkhangelsky et al., 2021). A visually large point can still be imprecise, and a small point can be economically meaningful when the outcome is narrow. The text therefore reports the coefficient and unit-clustered standard error alongside the figure rather than asking the chart to carry the whole inference (Hsieh and Moretti, 2019; Goldsmith-Pinkham et al., 2020).

The example is the preferred event-time estimate. At event time 1, the coefficient is -36.812 and the standard error is 4.753. That number is read as a difference in log claimant count for high-exposure units relative to lower-exposure units, net of the fixed-effects and baseline-control structure described above (Borusyak et al., 2022; Adao et al., 2019).

The analysis starts with direction and then moves to precision (Rosenbaum, 2002; Kitchin, 2014). If the post-period coefficients sit consistently on one side of zero, the result points to an organized recovery pattern. If the confidence intervals are wide, the interpretation shifts toward the sign and timing of the pattern rather than a tightly estimated magnitude (Conley, 1999; Cameron et al., 2011).

The regression table gives the compact version of the same comparison (Sun and Abraham, 2021). The high-exposure by post-period coefficient is -33.366, with a unit-clustered standard error of 4.220. This coefficient is easier to compare across the four papers because each public-data panel reports the same estimand format while retaining its own outcome and controls (Angrist and Pischke, 2009; Wooldridge, 2010).

Table 2 also reports the continuous-exposure coefficient (Chodorow-Reich, 2019; Imbens and Rubin, 2015). Instead of cutting the exposure at the median, it uses the pre-period value itself. Agreement between the median-split and continuous rows strengthens the interpretation because the result is not driven only by the threshold. Disagreement is informative too, because it says the relationship may be nonlinear (Imbens and Rubin, 2015; Rosenbaum, 2002).

The robustness outcome prevents the paper from over-reading one measurement choice (Imbens and Rubin, 2015; for National Statistics, 2026a). In this paper the robustness margin is claimant-count annual change. A similar coefficient supports a broader interpretation of the mechanism; a weaker or opposite coefficient confines the conclusion to log claimant count. Either way, the paper learns something more precise than a binary pass-fail result (Manski, 2003; Rubin, 1974).

The controls are part of the result. When population density, education, or baseline labour-market measures are included, the coefficient is no longer a raw comparison between advantaged and disadvantaged places. It is a within-unit post-period contrast that allows standard baseline characteristics to have their own post-shock association with the outcome (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021).

The criticism to keep in view is residual selection (Kline and Moretti, 2014; Gelman et al., 2020). A controlled model can still miss changing sectoral composition, housing constraints, commuting spillovers, or local policy responses. Table 2 therefore interprets coefficients and confidence intervals as evidence about the measured recovery margin rather than as claims about every unobserved channel (Goodman-Bacon, 2021; Arkhangelsky et al., 2021).

The relation back to the question is direct (Gelman et al., 2020; for National Statistics, 2026b). Do local authorities with higher claimant exposure show different claimant-count persistence after 2020? The figure answers the timing version of that question, the regression table answers the average post-period version, and the robustness table asks whether the answer is confined to one outcome definition. That structure is why these papers are distinct from one another (Abadie et al., 2010; Athey and Imbens, 2016).

The results also create a reviewer-facing audit path (Angrist and Pischke, 2009). A reader can move from the summary table to the event figure, from the event figure to the regression table, and from the regression table to the robustness CSV. Each object has pretty labels and a named source, so the empirical claim is inspectable at the table level and not only in prose (Chernozhukov et al., 2018; Wager and Athey, 2018).

The coefficient should also be read against the baseline outcome (Neumark and Simpson, 2015; Manski, 2003). A movement in log claimant count can be small in table units while still mattering for local adjustment if the outcome is measured as a rate, growth change, or survey percentage. The paper therefore reports the sign, uncertainty, and sample together instead of treating statistical significance as the only result (Athey and Wager, 2019; Mullainathan and Spiess, 2017).

The event-study shape matters because recovery is not a single date. A post-period coefficient that appears immediately after 2020 has a different interpretation from one that emerges gradually. The figure lets readers see whether the pattern looks like a shock response, a delayed adjustment, or a continuation of pre-period movement (Gentzkow et al., 2019; Grimmer and Stewart, 2013).

The continuous-exposure estimate helps with economic scale (Arkhangelsky et al., 2021). A median split is easy to interpret, but it can hide how much exposure changes across units. The continuous row asks whether additional pre-period exposure is associated with additional movement in log claimant count. That gradient is especially useful when the high-exposure group contains places with very different baseline levels (Gelman and Hill, 2007; Gelman et al., 2020).

The robustness result is not a second headline (Moretti, 2011; Rosenbaum, 2002). It is a check on mechanism. If claimant-count annual change responds differently from log claimant count, the paper should say so and narrow the interpretation. If it moves similarly, the mechanism becomes more credible because the result is visible beyond one constructed dependent variable (Kitchin, 2014; Borgman, 2015).

Table 2 gives the control interactions a practical reading (Rosenbaum, 2002; Kitchin, 2014). They allow high-density, high-education, or stronger-baseline labour-market places to have different post-period slopes even when they are not high exposure on the main margin. That matters because the question is about pre-2020 claimant-count exposure, not about whether advantaged places generally recovered faster (Anselin, 1988; LeSage and Pace, 2009).

The final result is therefore a layered interpretation (Sun and Abraham, 2021). The figure gives timing, the table gives magnitude, the continuous row gives scale, the robustness row gives scope, and the controls make the comparison more credible. Those layers turn a seed topic into a paper that can be read, challenged, and improved without relying on generic recovery language (Openshaw, 1984; Fotheringham et al., 2002).

6 Discussion

6.1 Interpretation

The interpretation is substantive rather than apologetic (Greenstone et al., 2010). claimant exposure captures the local welfare and job-search margin before the shock and the post-2020 estimates show how that capacity margin is associated with the recovery path. The point is to explain the economic pattern visible in the table, not to recite every stronger design the paper lacks (Abadie et al., 2010; Athey and Imbens, 2016).

The result is strongest as a dataset-specific recovery pattern (Neumark and Simpson, 2015; Manski, 2003). In this paper, the relevant margin is log claimant count; the comparison is high versus lower pre-2020 claimant-count exposure; and the controls are pre-2020 employment rate, pre-2020 unemployment rate, NVQ4 qualification share. That combination is enough to make a separate empirical claim, while still leaving policy assignment and spillovers for a design with sharper treatment timing (Chernozhukov et al., 2018; Wager and Athey, 2018).

The most important threat is selection into exposure (Kitchin, 2014). Units with high pre-2020 claimant-count exposure may also have different institutions, sectoral composition, housing markets, or labour-force characteristics. The baseline controls reduce some of that pressure, but they do not turn the exposure into a randomized treatment. The better reading is comparative recovery evidence with transparent adjustment (Athey and Wager, 2019; Mullainathan and Spiess, 2017).

Spillovers also matter. Migration, employment, claimant counts, and establishment dynamics can move across local boundaries. A high-exposure unit may absorb activity from nearby lower-exposure units, or it may support neighbouring places through commuting and demand. The unit-year panel can reveal the pattern but not fully allocate cross-boundary gains (Gentzkow et al., 2019; Grimmer and Stewart, 2013).

6.2 Threats

The policy implication follows the measured mechanism (for National Statistics, 2026a). If labour-market distress persistence after the pandemic shock is the relevant channel, then recovery policy should measure that channel directly and collect the controls needed to separate it from baseline advantage. That means keeping population scale, density, education, and baseline labour-market status in the design whenever the outcome calls for them (Gelman and Hill, 2007; Gelman et al., 2020).

The next design should add explicit treatment timing rather than more caveats. Grant dates, eligibility thresholds, broadband roll-out, local restrictions, or programme exposure would let the same panel carry a clearer causal question. Until then, the evidence is best used to identify which local capacity margins deserve that sharper design (Kitchin, 2014; Borgman, 2015).

The issue in interpretation is how much weight to place on a public-data contrast (Busso et al., 2013; Imbens and Rubin, 2015). The answer depends on the pattern, not on a stock disclaimer. If the event path shifts after 2020, the static coefficient points in the same direction, and the robustness row is coherent, the paper has real evidence about a local adjustment margin (Moretti, 2011; Bartik, 1991; Kline and Moretti, 2014).

The example is the control set. Population density and education are not bureaucratic additions; they mark the kinds of local advantages that recovery studies repeatedly worry about. NOMIS labour-market controls play the same role for employment outcomes. Keeping them in the model makes the interpretation more economic and less title-driven (Pearl, 2009; Keele, 2015).

6.3 Policy Implications

The analysis also clarifies what the result is for (Austin et al., 2018; Gelman et al., 2020). It helps rank capacity margins that deserve further study, points to outcomes where public data already contain useful variation, and identifies where a sharper policy or treatment file would be worth joining. That is a productive conclusion, not a retreat from the empirical estimate (King et al., 1995; Shadish et al., 2002).

The criticism is that common shocks are blunt (Borgman, 2015). The same 2020 marker may cover different local exposure channels, from sectoral demand to commuting changes to administrative restrictions. The paper handles that by staying close to log claimant count and claimant-count annual change, while treating spillovers and treatment timing as the next design layer (Peng, 2011; Wilkinson et al., 2016).

The policy reading is therefore conditional (Moretti, 2011). If local capacity is visible in the measured recovery path, policy design should record both the capacity margin and the standard controls before the next shock. That means collecting density, education, baseline employment, unemployment, claimant exposure, and programme timing in the same local panel whenever the outcome requires them (for National Statistics, 2026b,a; Peng, 2011; Wilkinson et al., 2016).

6.4 Conclusion

The paper answers a separate question with a separate dataset: Do local authorities with higher claimant exposure show different claimant-count persistence after 2020? The evidence indicates a post-2020 difference in log claimant count by pre-2020 claimant-count exposure, estimated with unit and year fixed effects plus baseline controls.

The analytical reading is direct (Busso et al., 2013). The estimated path says where the recovery margin is visible, the controlled regression says how large the average post-period contrast is, and the robustness row says whether the interpretation travels to claimant-count annual change. That is stronger than a self-negating conclusion because it states what the evidence does show before describing what a later design could add.

The practical implication is that public data can support useful recovery papers when each one has its own question, source, controls, and outcome. This paper contributes one such object: a NOMIS Claimant Count panel that turns labour-market distress persistence after the pandemic shock into an inspectable empirical margin.

The contribution is also comparative across the four-paper run (for National Statistics, 2026a). Each paper now has its own public source, outcome, exposure, and literature-motivated controls. That matters because stronger production should not merely rewrite the same topic area; it should generate separate empirical objects that can pass reviewer gates on data, methods, and interpretation (for National Statistics, 2026a; Kitchin, 2014; Borgman, 2015).

The final reading is practical. The paper shows how to move from a seed idea to a reviewable public-data estimate: define the recovery margin, choose the dataset, motivate the controls, estimate the event path, report the compact coefficient, and interpret the mechanism without burying the result under self-negating caveats (Bureau, 2025; Wilkinson et al., 2016).

The same structure is reusable without making the papers repetitive. What repeats is the discipline: public source, motivated controls, fixed-effects comparison, robustness margin, and restrained interpretation. What changes is the empirical object: NOMIS Claimant Count, log claimant count, pre-2020 claimant-count exposure, and the mechanism that connects them (Bartik, 1991; Kline and Moretti, 2014).

A Data Construction

The data construction appendix reports the derived public panel and its provenance. Raw public-data payloads are hashed and discarded after the derived files are written (Peng, 2011; Wilkinson et al., 2016).

B Supplementary Source Evidence

The supplementary source evidence records the retrieval surface used for framing and reference checks. It keeps the literature sample, access rates, and search diagnostics visible without substituting them for the subject data (Priem et al., 2022; OpenAlex, 2026).

Table 3: Open-access literature sample

Quantity	Value
Retained open-access records	51
Records with direct document link	51
Document-access share	1.000
Publication-year cells	4

Table 4: Evidence-coverage regression estimates

Model	Outcome	Coef.	SE	95% CI	N
Annual open-access record trend	records per year	-3.100	3.357	[-9.680, 3.480]	4
PDF availability trend	direct PDF indicator	0.000	0.000	[0.000, 0.000]	51
Topic relevance and attention	log cited-by count	-5.764	3.940	[-13.486, 1.959]	51

Table 5: Robustness checks for the evidence-coverage trend

Sample	N	Trend	SE	95% CI
all records	51	-3.100	3.357	[-9.680, 3.480]
high relevance	1	0.000	0.000	[0.000, 0.000]
recent records	5	0.000	0.000	[0.000, 0.000]
PDF available	51	-3.100	3.357	[-9.680, 3.480]

C Supporting Regressions and Robustness

The supporting estimates expand the main result into the event-study table, robustness outcome, and pre-trend placebo. Baseline controls are interacted with the post period so the comparison is not carried by the exposure variable alone (Angrist and Pischke, 2009; Wooldridge, 2010).

Table 6: NOMIS Claimant Count event-study estimates

Model	Outcome	Event time	Coefficient	Std. error	95% CI	N
UK Local Authority Event Study	Log Claimant Count	-3	-1.902	3.397	[-8.560, 4.757]	2030
UK Local Authority Event Study	Log Claimant Count	-2	3.087	2.257	[-1.337, 7.510]	2030
UK Local Authority Event Study	Log Claimant Count	0	-14.968	1.995	[-18.878, -11.057]	2030
UK Local Authority Event Study	Log Claimant Count	1	-36.812	4.753	[-46.128, -27.495]	2030
UK Local Authority Event Study	Log Claimant Count	2	0.000	0.000	[0.000, 0.000]	2030
UK Local Authority Event Study	Log Claimant Count	3	0.000	0.000	[0.000, 0.000]	2030

Notes: Event time -1 is the omitted category. Estimates include unit and year fixed effects with unit-clustered standard errors.

Table 7: Public-panel robustness checks

Robustness check	Outcome	Coefficient	Estimate	Std. error	95% CI	N
Alternative Outcome: Claimant-count Annual Change	Claimant-count Annual Change	High Exposure by Post-2020	252.238	49.902	[154.430, 350.046]	2030
Pre-trend placebo	Log Claimant Count	High Exposure by Pre-period Trend	0.951	1.697	[-2.376, 4.278]	1218

Notes: Estimates include the fixed effects and clustered standard errors reported in the generated regression CSV.

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